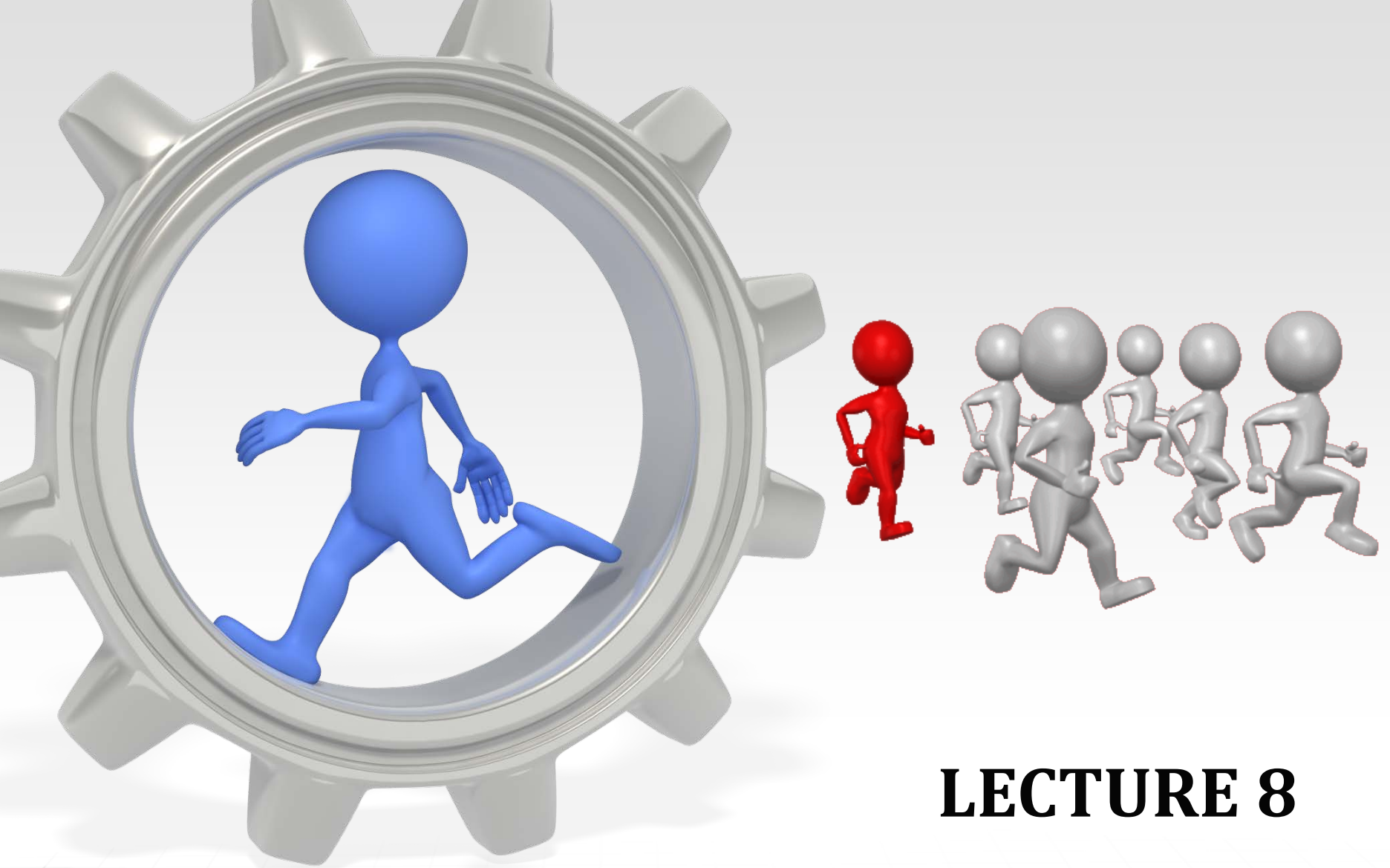




LECTURE 8

CONCURRENT ENGINEERING AND PROCESS PLANNING

Contents

1. Objectives
2. Review
3. Process planning
4. Computer aided process planning
5. Concurrent engineering and Design for manufacturing and assembly
6. Advanced manufacturing planning
7. Conclusion



Objectives

1. Specific definitions in planning processes.
2. Knowing the characteristics of computer-aided planning.
3. Understand the importance of concurrent engineering.
4. Understand the design for processing and assembly concept.
5. Identify specific activities for the advanced planning of manufacturing.



Review

1. Types of flexibility in manufacturing
2. The benefits of an FMS
3. Tests of flexibility (part variety, schedule change, error recovery, new part)





Process planning (1)

Process planning is the activity of developing a manufacturing plan used to convert the product design into a physical entity.

Developing a process plan involves:

- interpretation of design drawings,
- determining the processes and sequence,
- equipment selection,
- tools, dies, molds, fixtures, and gauges selection,
- analysis methods.
- work standards,
- cutting tools and cutting conditions, etc.



Process planning (2)

Process planning can be done:

- manually
- automated using computers (CAPP =Computer Aided Process Planning)

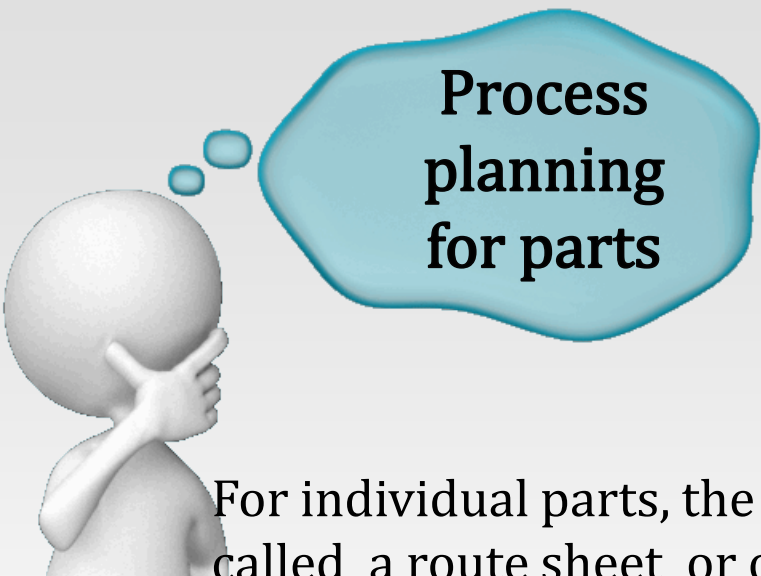
☐ Process plans developed by different people can be different for the same part or product .

☐ Process planning is necessary for:

- part produced,
- assembly



Consistent and correct planning requires both knowledge of the manufacturing processes and experience.

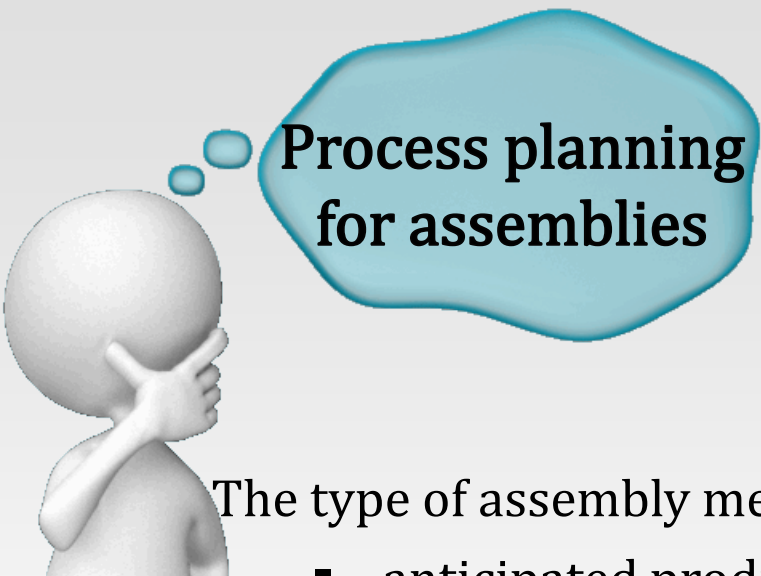


Process planning for parts



For individual parts, the processing sequence is documented on a form called a route sheet or operation sheet:

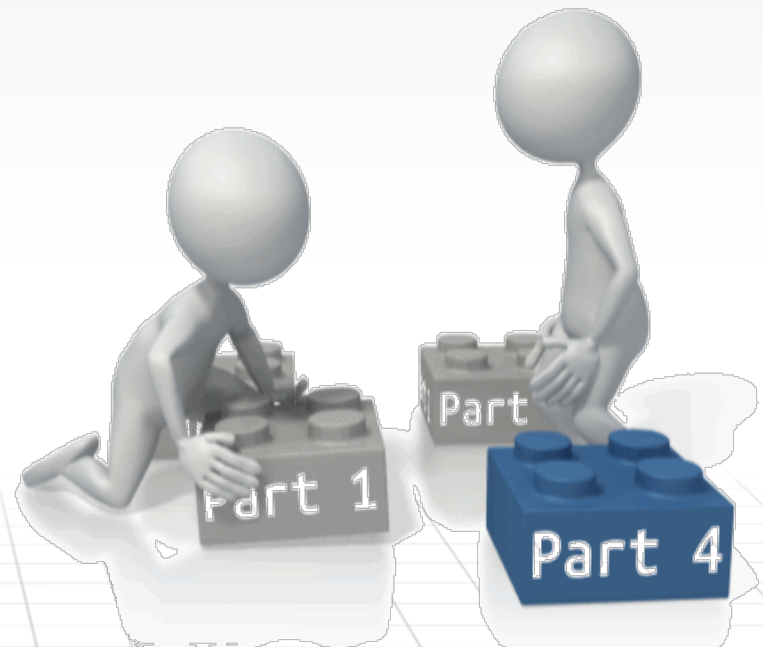
Route sheet		Metal Steel Company				
Part no 65352	Part name Shaft	Planner Popescu I	Checked by: Ionescu P	Date 08/12/08	Page 1/1	
Material 1050 H18 Al	Stock size Ø 60 mm, 206 mm length	Comments				
No.	Operation description	Dept	Machine	Tooling	Setup	Std
10	<i>Face end (approx. 3 mm) Rough turn to Ø 52.00 mm. Finish turn to Ø 50.00 mm.</i>	<i>Lathe</i>	<i>L45</i>	<i>G0810</i>	<i>1.0hr</i>	<i>5.2 min</i>
20	<i>Drill 4 radial holes Ø 7.50 mm</i>	<i>Drill</i>	<i>D09</i>	<i>J555</i>	<i>0.5 hr</i>	<i>3.2 min</i>
---	-----	---	---	---	---	---
50	<i>Mill 6.5 mm deep x 5.00 mm wide slot</i>	<i>Mill</i>	<i>M13</i>	<i>F630</i>	<i>0.7 hr</i>	<i>4.8 min</i>

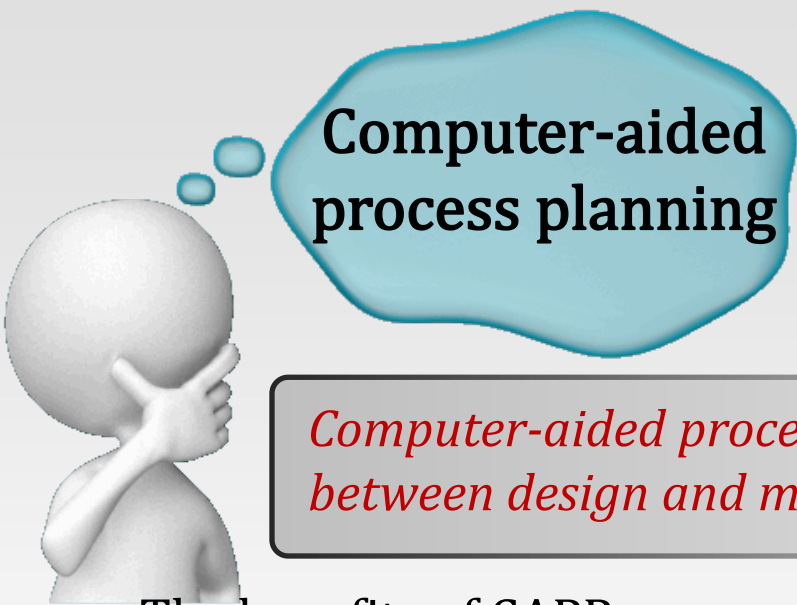


Process planning for assemblies

The type of assembly method used for a given product depends on:

- anticipated production quantities,
- complexity of the assembled product,
- assembly processes used:
 - mechanical assembly,
 - welding





Computer-aided process planning

Computer-aided process planning (CAPP) is the direct connection between design and manufacturing.

The benefits of CAPP are:

- process rationalization and standardization
- increased productivity of process planners
- reduced lead time for process planning
- improved legibility
- incorporation of other application program

CAPP systems are designed around three approaches:

- retrieval CAPP systems
- generative CAPP systems
- semi-generative CAPP systems



Retrieval CAPP systems (1)

A retrieval CAPP system, also called a variant CAPP system, is based on the principles of group technology and parts classification and coding.

The development of the data base of these process plans consists of the following steps:

- **selecting** an appropriate classification and coding scheme for the company
- **forming** part families for the parts produced by the company
- **preparing** standard process plans for the part families

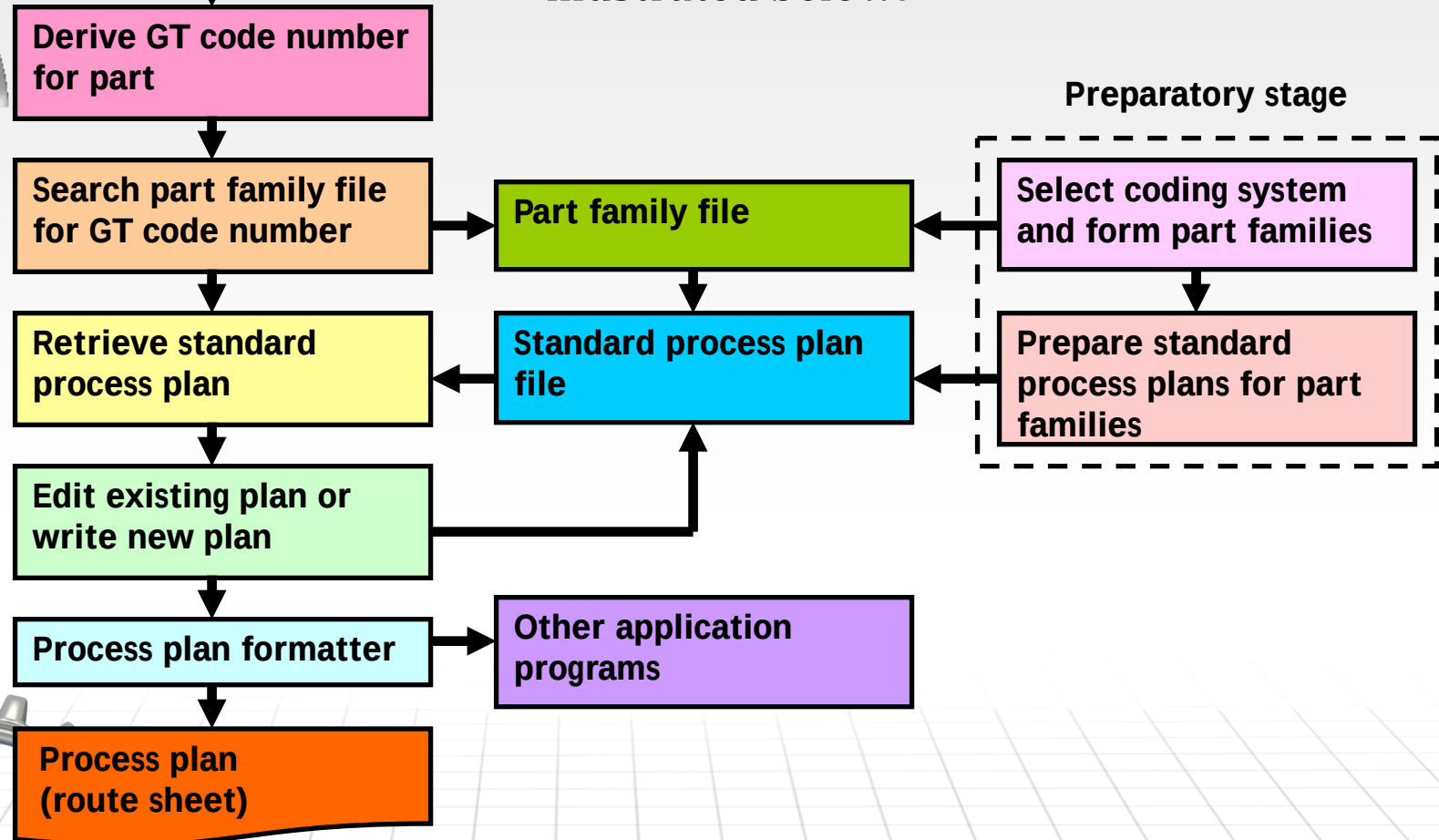


The standard route sheets are based on current part routings in use in the factory or on an ideal process plan that has been prepared for each family.

Retrieval CAPP systems (2)

New part design

The retrieval CAPP system operates as illustrated below:





Generative CAPP systems (1)

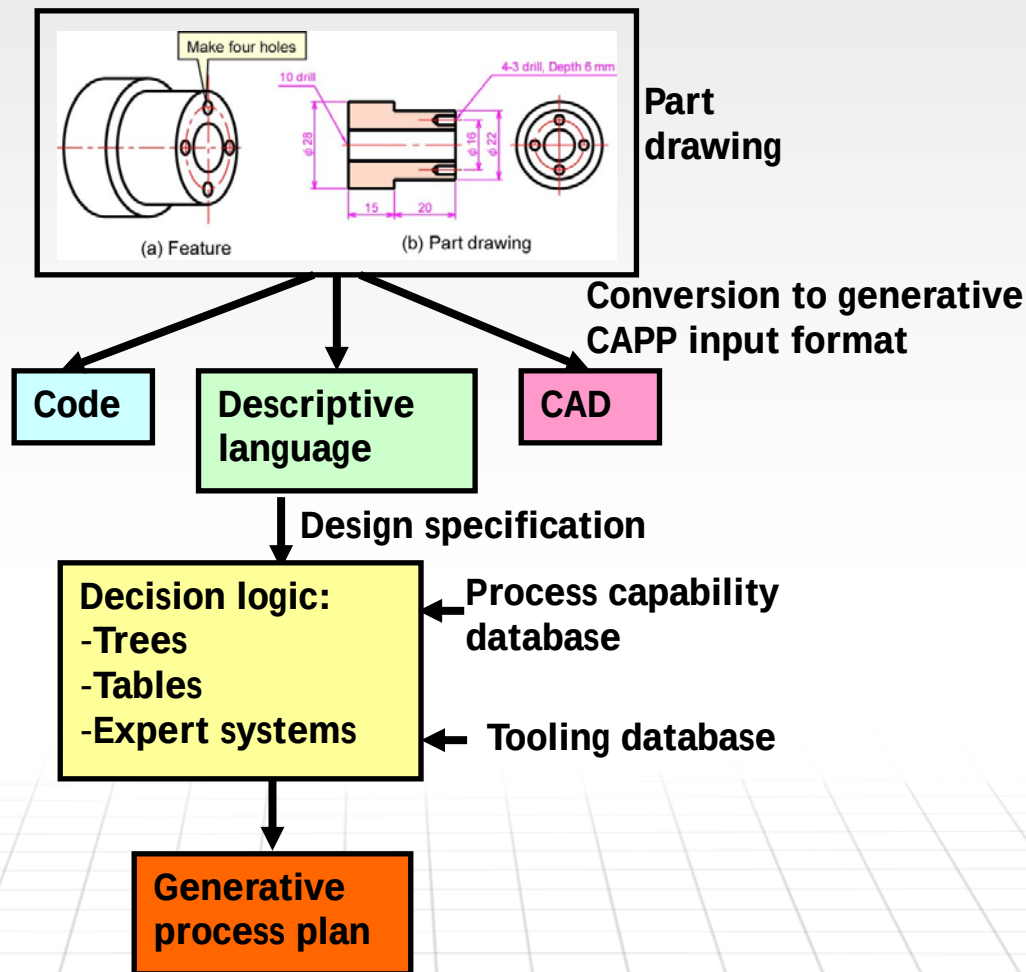
- ❑ The CAPP generative technique is both more difficult to develop and more highly automated.
- ❑ The generative process planning system creates plans for new components without referring to existing plans or with the assistance of a human planner.
- ❑ The heuristic planning knowledge is captured in artificial intelligence (AI) software, called expert systems.

The advantages of generative CAPP :

- process plans are created rapidly and consistently;
- totally new process plans are created as quickly as plans similar to those for existing components;
- an interface to MRP and ERP software is possible;
- documentation requirements of ISO or QS 9000 standards are completed and best practices are promoted, even without experienced workers.

Generative CAPP systems (2)

The generalized system illustrated below shows the generative CAPP operation:





Self-assessment test

1. Processes Planning requires information on existing tools and fasteners.

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2. Processes Planning is always made automatically in a company.

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3. PP requires information about manufacturing processes and experience to achieve a consistent planning.

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4. The assembling method of a product depends on the expected throughput.

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5. Use of a computer in planning processes decreases the time required to develop the process plan.

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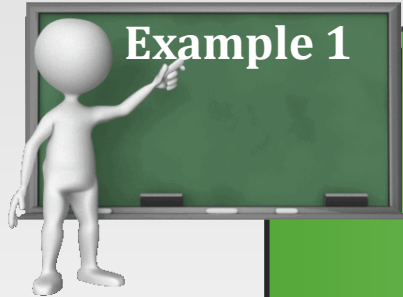


Make or buy ?

There are a number of factors that enter into the make or buy decision:

- 
- ☐ How do part costs compare?
 - ☐ Is the process available in-house?
 - ☐ What is the total production quantity?
 - ☐ What is the anticipated product life?
 - ☐ Is the component a standard item?
 - ☐ Is the supplier reliable?
 - ☐ Is the company's plant already operating at full capacity?
 - ☐ Does the company need an alternative supply source?

Make or buy ?



Example 1

The quoted price for a certain part is 20 euros per unit for 100 units. The part can be produced in the company's own plant for 28 euros. The cost components of making the part are listed below.

Components	Price/unit [euro]
Unit raw material cost	8
Direct labor cost	6
Labor overhead at 150%	9
Equipment fixed cost	5
Total	28

Should the component be bought or made in-house?

Solution

What is the impact on plant if the quote is accepted?

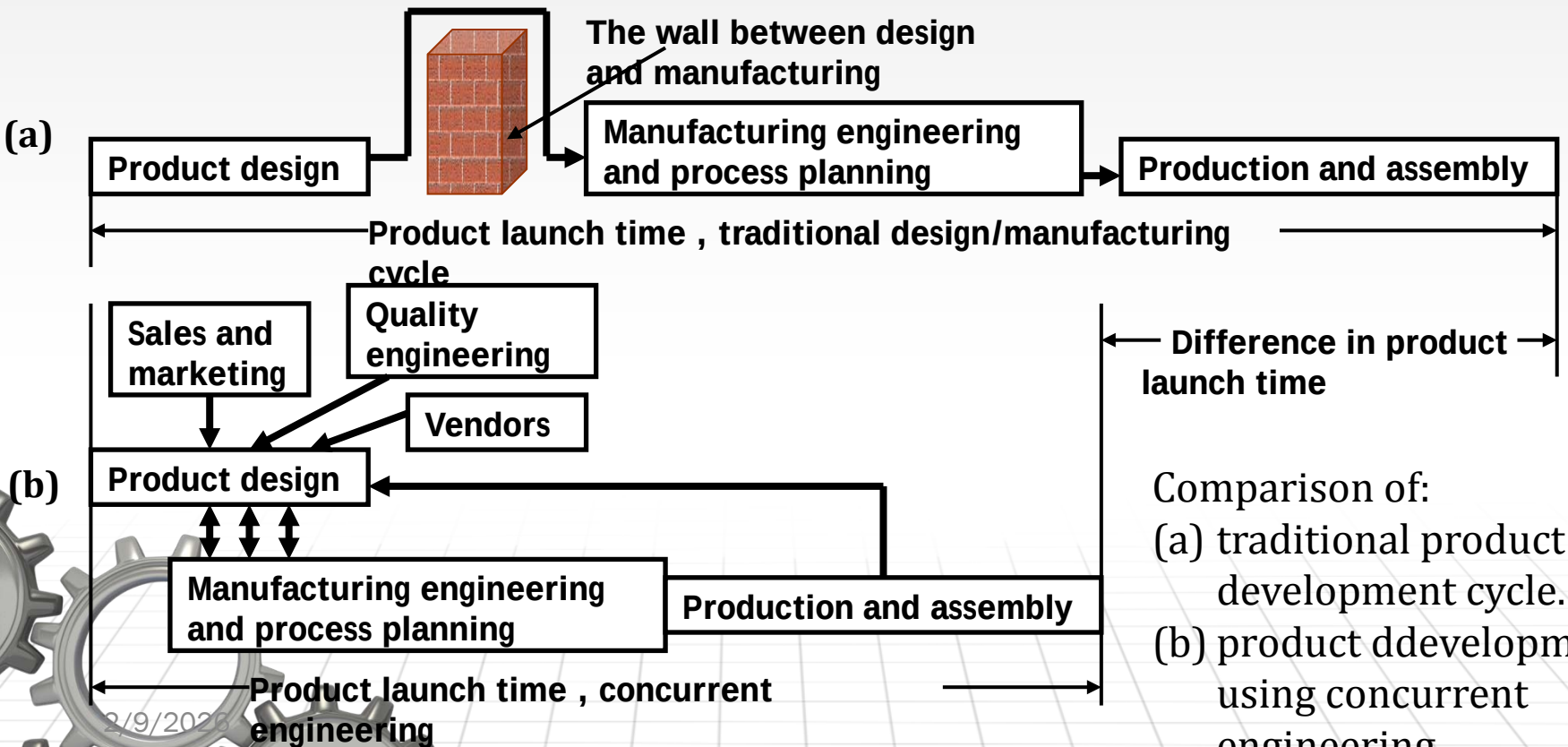
The cost components for the company, if the decision is to buy the part are listed below:

Components	Price/unit [euro]
Quoted price for one part	20
Labor overhead at 150%	9
Equipment fixed cost	5
Total	34

Make or buy ?

Concurrent engineering (1)

Concurrent engineering, also called simultaneous engineering refers to an approach used in product development in which the functions of design engineering, manufacturing engineering, and other functions are integrated to reduce the elapsed time required to bring a new product to market.





Concurrent engineering (2)

- ❑ Concurrent engineering includes several elements:
 - design for manufacturing and assembly
 - design for quality
 - design for cost
 - design for life cycle

- ❑ Additionally certain technologies are required to facilitate the concurrent engineering:
 - rapid prototyping
 - virtual prototyping
 - organizational changes



Design for manufacturing and assembly

Design for manufacturing and assembly (DFMA) is any procedure or design process that considers the production factors from the beginning of the product design.

❑ DFMA provides the following typical advantages:

- Reduction of design-to-product launch cycles by as much as 50 percent.
- Reduction of part counts by 30 to 70 percent.
- Reduction of assembly times by 50 to 80 percent.

❑ Design for manufacturing and assembly involves:

- organizational changes
- design principles and guidelines.



Organizational changes in DFMA

❑ Effective implementation of DFMA involves making changes in a company's organizational structure:

- by creating project teams consisting of:

- ✓ product designers
- ✓ manufacturing engineers
- ✓ quality engineers
- ✓ material scientists

- by requiring design engineers to spend some time in manufacturing;

- by assigning manufacturing engineers to the product design department to serve as manufacturing consultants.





Design principles and guidelines

- 
- 
- 
- ❑ General principles and guidelines in DFMA are:
 - minimize number of components;
 - use standard commercially available components;
 - use common parts across product lines;
 - design for ease of part fabrication;
 - design parts with tolerances that are within process capability;
 - design the product to be foolproof during assembly;
 - minimize flexible components;
 - design for ease of assembly;
 - use modular design;
 - shape parts and products for ease of packaging;
 - eliminate or reduce adjustments.



Design for quality

Design for quality (DFQ) is the term that refers to the principles and procedures employed to ensure that the highest possible quality is designed into the product.

❑ The general objectives of DFQ are:

- to design the product to meet or exceed customer requirements
- to design the product so that its function and performance are relatively insensitive to variations in manufacturing and subsequent application.
- to continuously improve the performance, functionality, reliability, safety, and other quality aspects.





Design for product cost

Design for product cost (DFC) refers to the efforts of a company to specifically identify how design decisions affect product costs and to develop ways to reduce cost through design.



Design for life cycle

Design for life cycle (DFLC) refers to the product after it has been manufactured and includes factors ranging from product delivery to product disposal.



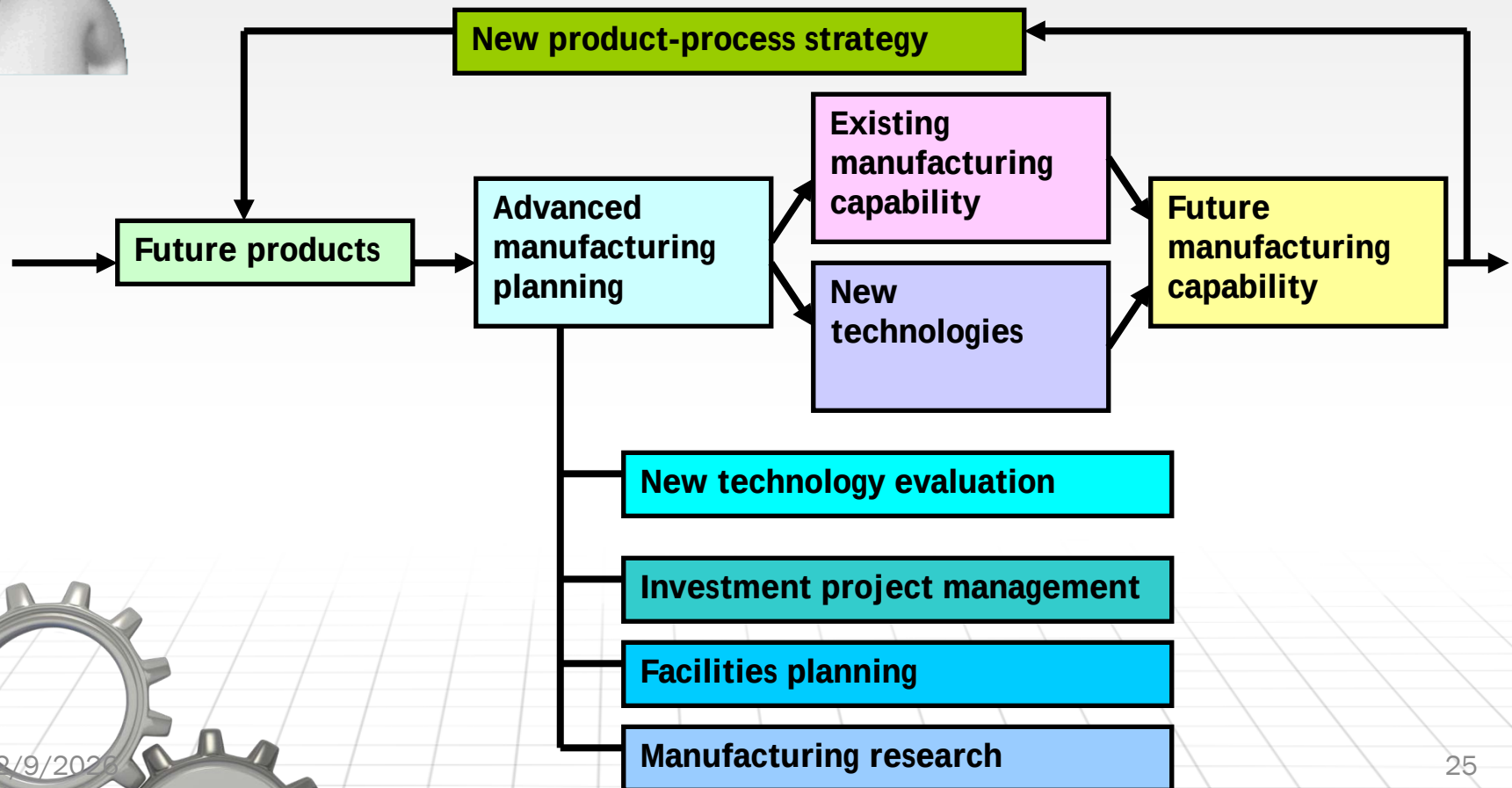
Advanced manufacturing planning (1)

Advanced manufacturing planning involves working with sales, marketing, and design engineering to forecast the new products that will be introduced and to determine what production resources will be needed to make those future products.

- ❑ Activities in advanced manufacturing planning include:
 - new technology evaluation,
 - investment project management,
 - facilities planning,
 - manufacturing research.

Advanced manufacturing planning (2)

The general planning cycle:





New technology evaluation

- ❑ To introduce a new product a company must either:
 - implement new processing technologies in-house or,
 - purchase the components made by the new technologies.

- ❑ Other reasons why a company may need to introduce new technologies:
 - quality improvement
 - productivity improvement
 - cost reduction
 - lead time reduction
 - modernization and replacement of worn-out equipment



Investment project management



- ❑ The steps for investment project management activity are:
 - proposal to justify the investment is prepared;
 - management approvals are granted for the investment;
 - vendor quotations are solicited;
 - order is placed to the winning vendor;
 - vendor progress in building the equipment is monitored;
 - any special tooling and supplies are ordered;
 - the equipment is installed and debugged;
 - training of operators;
 - responsibility for running the equipment is turned over to the operating department.



Facilities planning

Facilities planning is the planning work required to renovate an existing facility or design a new one, and is carried out by the plant engineering department.

- ❑ Facilities planning can be divided into two types of problems:
 - facilities location
 - facilities design



Factors that must be considered: location relative to customers and suppliers, labor availability, skills of labor pool, transportation, etc..



Manufacturing research and development

- ❑ Manufacturing research can take various forms:
 - development of new processing technologies,
 - adaptation of existing processing technologies,
 - process fine-tuning,
 - software systems development,
 - automation systems development,
 - operations research and simulation.





Self-assessment test

1. Retrieval CAPP systems use the principles of group technology.

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2. Generative CAPP systems use artificial intelligence algorithms.

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3. Using concurrent engineering concepts, the lead time decreases.

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4. Design for manufacturing and assembly involves changes at the organizational level in a company.

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5. Manufacturing research involves the optimization of production processes.

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Programs packages used in CAPP

Examples of companies that develop and sale CAPP software:

- **CIMx Company → CIMx Apps® Suite**

(<http://www.cimx.com/solutions-and-benefits>)

- **MTI SYSTEMS → Costimator OEM, COSTIMATOR JS**

(<http://www.mtisystems.com/>)



Conclusion



- ☐ Production processes planning involves the use of information on production technologies.
- ☐ Have to take into account aspects of the structure of production, the equipment available, qualification of personnel involved, etc..
- ☐ Proper planning of production processes is a key factor in ensuring high productivity.





- ☐ Production Planning
- ☐ Material Requirements Planning
- ☐ Capacity Planning